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Department of Artificial Intelligence and Data Science

VIRO - AI

Viral Insight & Rapid Optimization - Analytics Intelligence

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Year : TE

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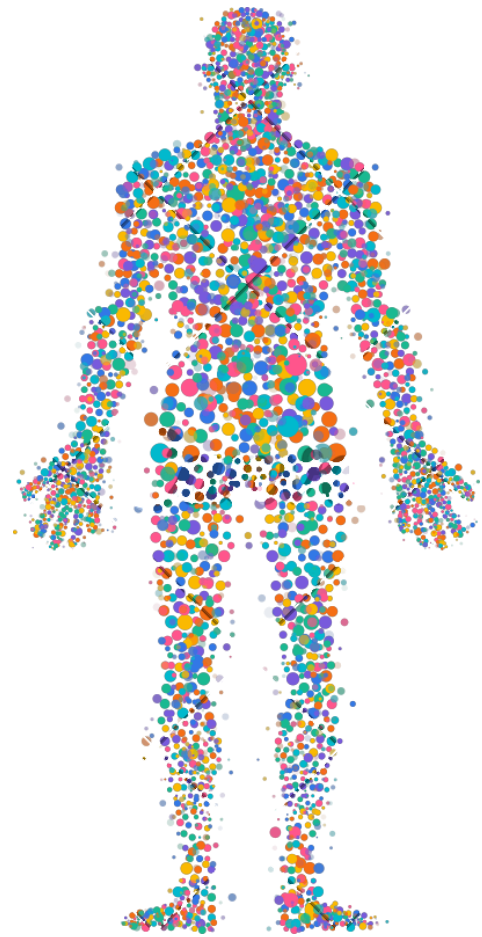
Stream : Artificial Intelligence & Data Science

Year : TE





INTRODUCTION



- **Project Name :** VIRO - AI : Viral Insight & Rapid Optimization Analytics Intelligence
- **Domain :** MedTech | BioTech | HealthTech
- **Category :** Software
- **Core Goal :** Predict viral mutations, recommend drug candidates, and accelerate drug discovery.
- **Approach :** Combines AI/ML, bioinformatics, and 3D structural modeling for proactive pandemic response.
- **Team Name :** Rakshanu



PROBLEM STATEMENT



Cutting-edge technology in these sectors continues to be in demand. Recent shifts in healthcare trends, growing populations also present an array of opportunities for innovation.

Unpredictable Viral Mutations

Current systems are slow to track and predict mutations.

Delayed Drug Discovery

Traditional lab-based drug testing takes months/years.

Reactive Healthcare

Outbreaks detected only after spread, no early-warning system.

Lack of Context

Viral prediction ignores real-world vitals/environmental factors.

Viro-AI is a powerful, **AI-driven** platform that predicts viral mutations and suggests potential drug candidates to accelerate outbreak response and research.



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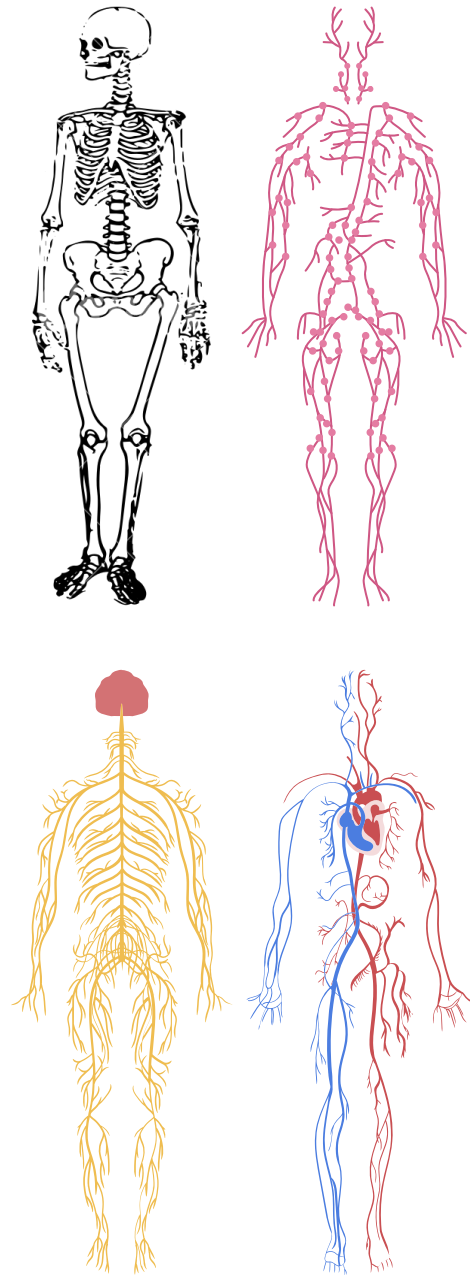
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PROPOSED SOLUTION



Viral Insight & Rapid Optimization Analytics Intelligence



Mutation Prediction

AI analyzes genome and protein data to identify and highlight potential viral mutations.

Drug Candidate Suggestion

A cheminformatics engine evaluates and ranks promising antiviral compounds for treatment.

Dashboard Visualization

Interactive dashboards display mutation hotspots, candidate drugs, 3D protein structures, and associated risk indices.

Proactive Threat Management

Insights from **data-driven** predictions help healthcare systems prepare and respond to emerging viral threats effectively.



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TECHNICAL APPROACH



Tech & Tools

- **Databases**
 - SQLite, PostgreSQL, Parquet, HDF5
- **Containerization**
 - Docker
- **Monitoring**
 - Prometheus, Grafana
- **Experiment Tracking**
 - MLflow, Optuna

Libraries

- **Data Handling**
 - Pandas, NumPy, Dask, Apache Spark
- **ML/AI**
 - Scikit-learn, TensorFlow/Keras, PyTorch, XGBoost, LightGBM
- **Bioinformatics**
 - BioPython, RDKit, MDAnalysis
- **3D Visualization**
 - PyVista, Plotly, PyMOL, Open3D
- **Web/API**
 - Flask, FastAPI, Streamlit, Dash

Technical Language

- **Python**
 - Backend Development
 - AI Model Implementation
 - NPL
- **JavaScript**
 - Frontend Development
 - Handling User Interactions
- **HTML / CSS**
 - Frontend Development
 - Building User Interface
- **SQL**
 - Database Management

Algorithms

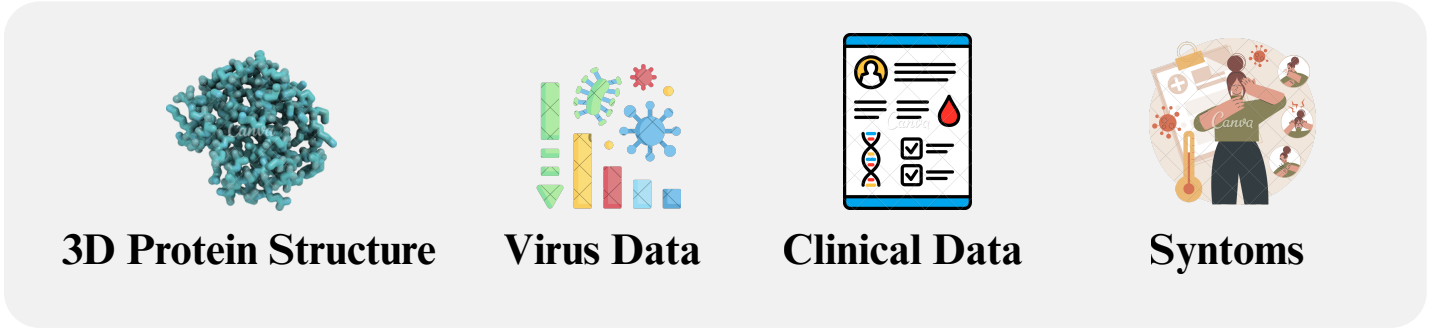
- **Data Processing**
 - ETL pipelines, streaming algorithms
- **Mutation Prediction**
 - Random Forest, Gradient Boosting, LSTM/GRU, CNN
- **Drug Discovery**
 - Molecular docking (AutoDock/Vina), QSAR models, similarity search (Tanimoto)
- **Visualization & Simulation**
 - Molecular dynamics, 3D rendering, mesh processing



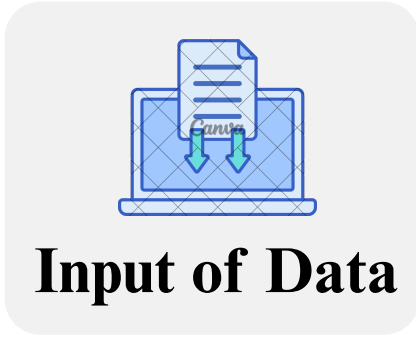
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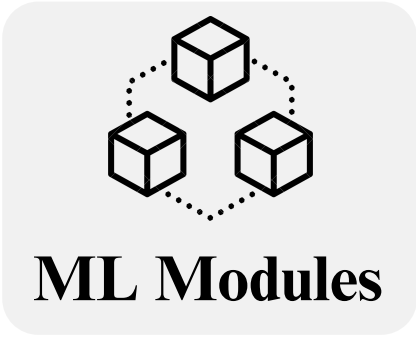
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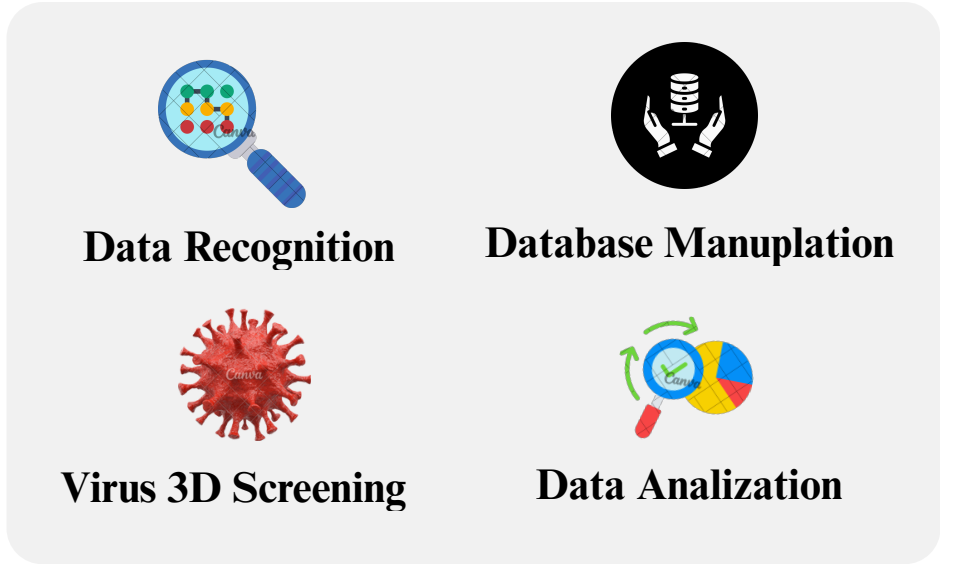
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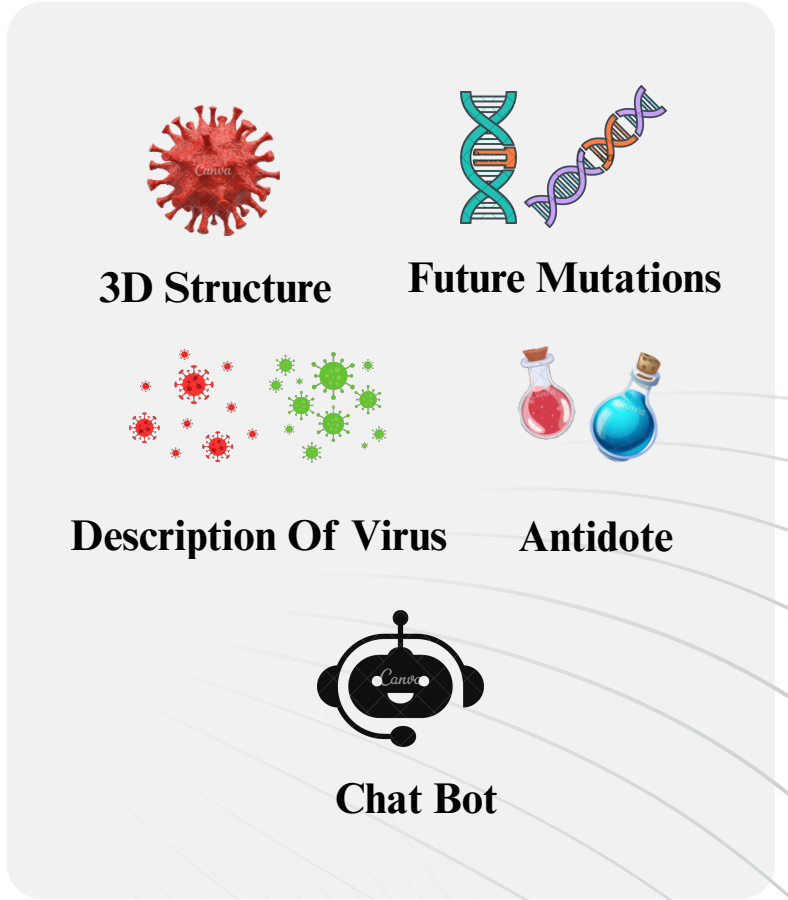
Processing



Output



Add the new virus data to our database



SYSTEM ARCHITECTURE



KEY FEATURES



Multi-Modal AI

The system combines genomic, structural, and text data using an ensemble of AI models for highly accurate predictions.

End-to-End System Integration

Viro-AI automates viral mutation prediction, drug screening, and 3D analysis.

Accelerated Screening

The system uses AI to perform parallel screening of over 10,000 compounds, dramatically reducing the time needed to find potential antidotes.

Rapid Threat Assessment

It generates a custom "Deadliness Score" with a confidence interval, providing a quick, quantifiable measure of a virus's threat level.

Architecture

Built on a scalable, cloud-based architecture with open-source technologies.

Interactive 3D Visualization

The prototype includes real-time, interactive 3D visualizations and simulations to bring complex molecular interactions to life.



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FEASIBILITY & VIABILITY



Strengths & Readiness

Architecture

Built on a scalable, cloud-based architecture with open-source technologies.

Data Strategy

Leverages a continuous stream of public data for model training and validation

Production Readiness

Emphasizes a maintainable structure for large-scale, real-world deployment.

Potential Challenges

Regulatory Compliance

Navigating global health data regulations is a critical challenge.

Trust & Transparency

Risk of algorithmic bias could erode trust; requires model explainability.

Operational Costs

Managing high data volume and requests efficiently while being cost-effective.

Strategies

Security

Implement a proactive security framework with encryption and access controls.

Explainability

Integrate Explainable AI (XAI) to provide transparent reasoning for decisions.

Scalability

Use dynamic cloud infrastructure with containerization to optimize performance and costs.



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IMPACT & BENEFITS



Benefits

Social

Early detection and faster response save lives during outbreaks.

Economic

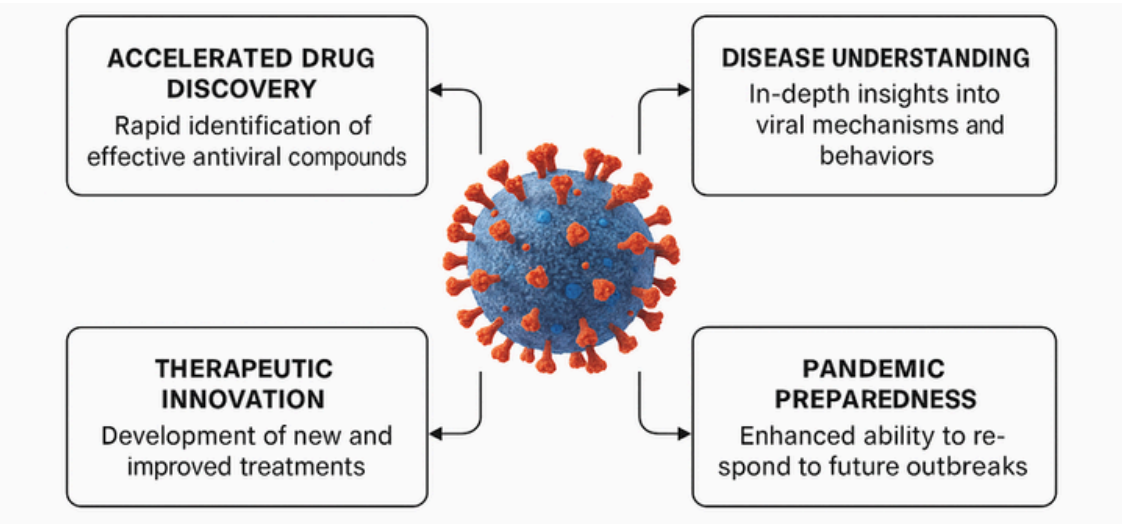
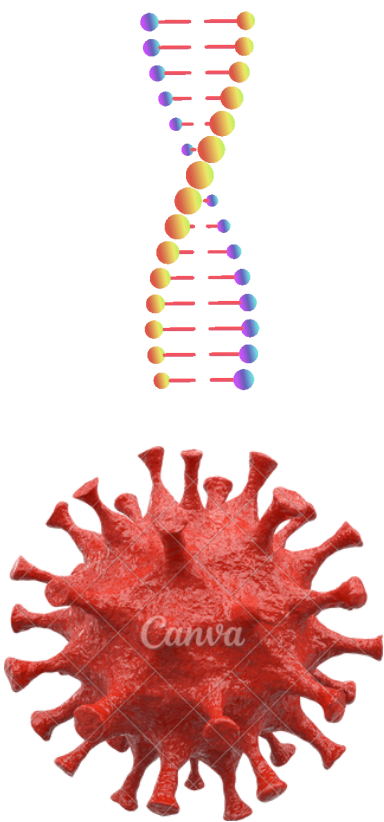
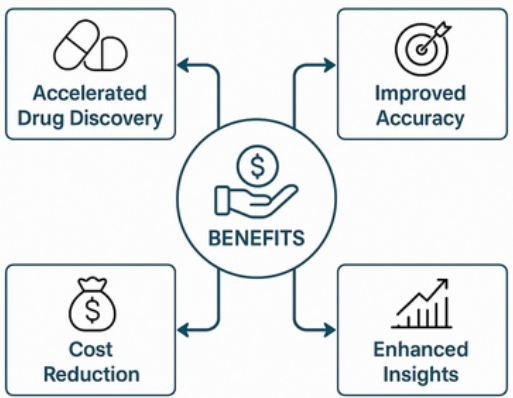
Cuts down cost and time in drug discovery using AI + Cheminformatics.

Healthcare

Provides clear visual insights to support medical professionals.

Technological

Integrates AI, ML, Cheminformatics, and 3D Visualization into one solution.



Impact

Accelerated Research

AI/ML models speed up genome and protein analysis, reducing months of work to hours.

Proactive Response

Predicting mutation hotspots enables faster vaccines and treatments.

Healthcare Support

Dashboard simplifies bioinformatics, helping labs and hospitals take timely decisions.

Global Preparedness

Scalable system with potential for LLM-based biomedical reasoning to improve pandemic readiness.



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BUSINESS IDEOLOGY



Phase 1

Research & Validation (Years 1-2)

- **Focus :** Establish Scientific Credibility
- **Product :** Research-as-a-Service (RaaS).
- **Target Audience :** University Labs & Biotech Startups.
- **Revenue Model :** Research Grants & Project-Based Contracts.
- **Key Goal :** Achieve >70% prediction accuracy in a published case study and secure foundational research partners.

Phase 2

Commercialization (Years 2-4)

- **Focus :** Build a Scalable Business
- **Product :** Tiered SaaS Platform (Basic, Pro, Enterprise).
- **Target Audience :** Mid-Sized Pharma, CROs, and Health Agencies.
- **Revenue Model :** Recurring Subscriptions & Usage-Based Fees.
- **Key Goal :** Onboard first paying enterprise clients and achieve key regulatory compliance (e.g., HIPAA).

Phase 3

Ecosystem Expansion (Year 5+)

- **Focus :** Become an Industry Standard
- **Product :** Predictive Health Intelligence Platform.
- **Target Audience :** Large Pharma, Governments, and Global Health Organizations (WHO).
- **Revenue Model :** Government Contracts & API Marketplace Revenue Sharing.
- **Key Goal :** Integrate into national pandemic preparedness plans.



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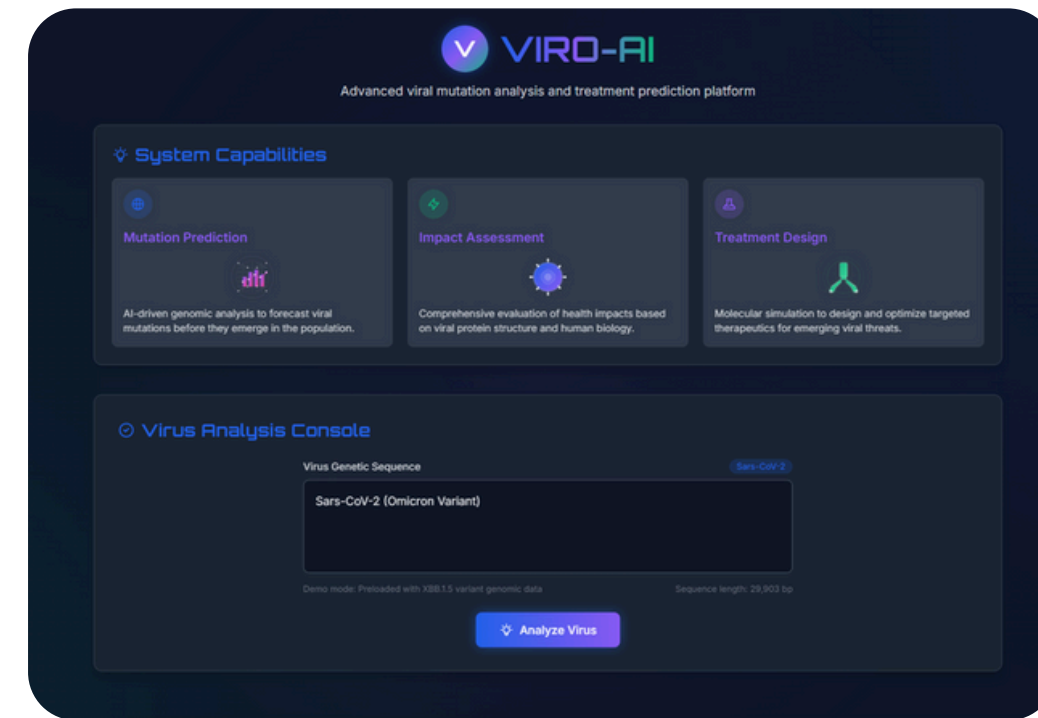
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PROTOTYPE



- Predictive Analytics
- Dynamic Visualization.
- Molecular Simulation
- Actionable Insights.



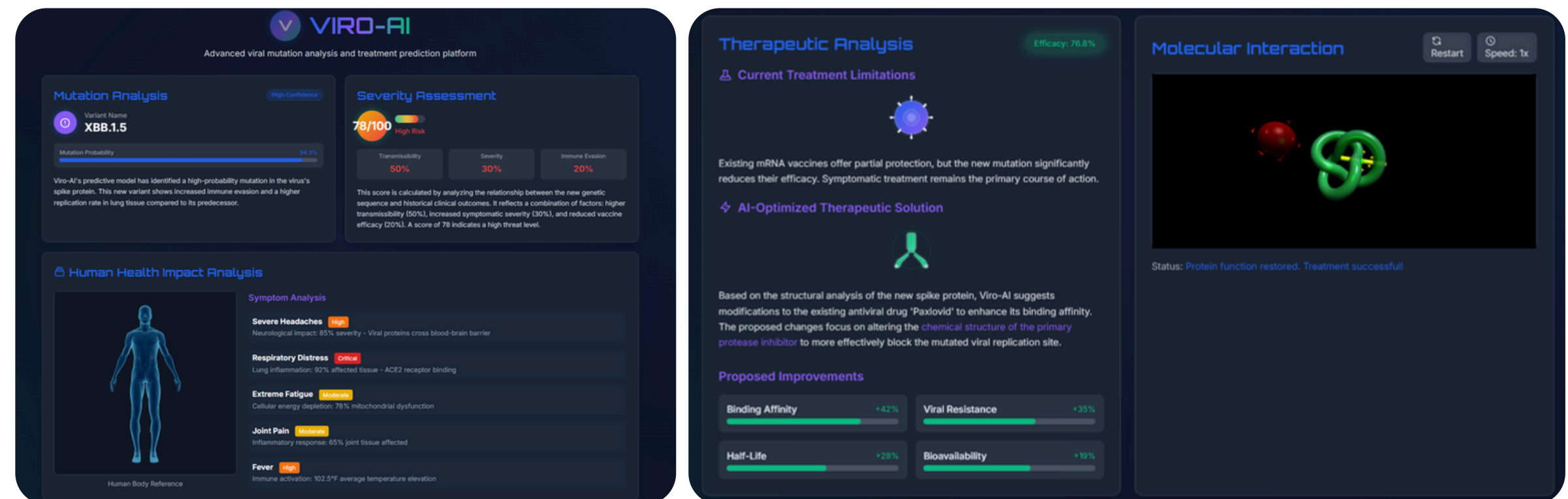
Page 1 : Data Input



Scan for Real Time
Prototype Experience



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Page 2 : Result Dashboard



RESEARCH & REFERENCES



Sr. No	Author	Paper Title	Publication	Year	Link
1)	Jumper, J. et al.	AlphaFold: Highly accurate protein structure prediction	Nature	2021	https://www.nature.com/articles/s41586-021-03819-2
2)	Kokudeva, M. et al.	AI as a tool in drug discovery and development	Biotechnology & Biotechnological Equipment	2021	https://www.nature.com/articles/s41573-019-0016-z
3)	Deng, J. et al.	Artificial intelligence in drug discovery: applications & techniques	Briefings in Bioinformatics	2020	https://www.nature.com/articles/s41573-019-0016-z
4)	Vamathevan, J. et al.	Machine learning in drug discovery and development	Nature Reviews Drug Discovery	2019	https://www.nature.com/articles/s41573-019-0016-z



“For more details about this project, please scan the QR code to access a detailed overview.”



CONCLUSION



A New Era of Proactive Health

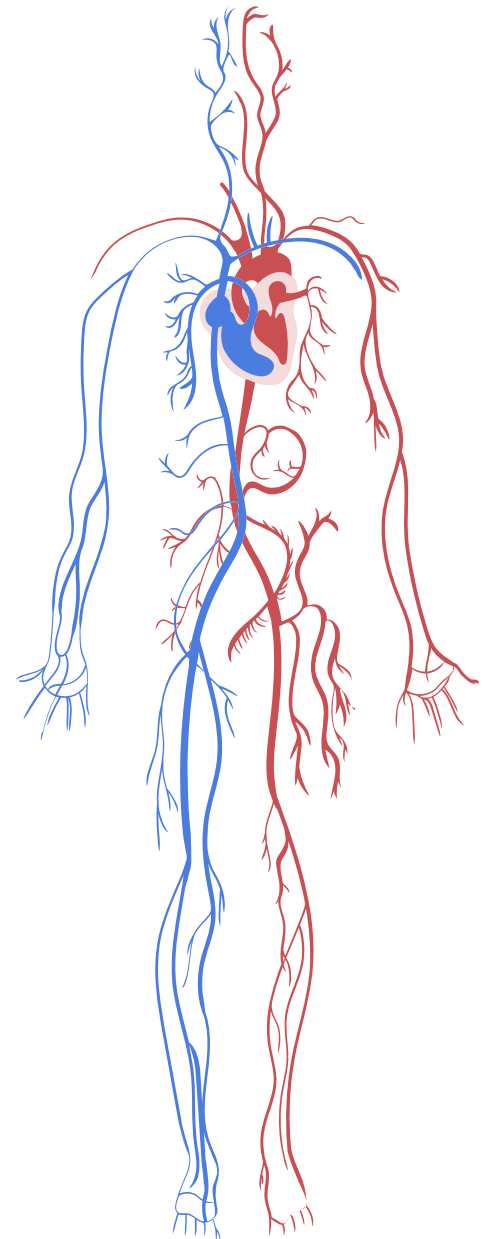
Viro-AI is not just a tool—it's a paradigm shift in our fight against infectious diseases.

Accelerate Discovery : We dramatically reduce the time and cost of developing life-saving treatments.

Empower Researchers : We provide the data-driven foresight needed to stay ahead of viral evolution.

Secure Global Health : Our ultimate vision is a global, AI-powered defense system to prevent future pandemics.

Join us in building a healthier, safer future.



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ANY QUESTIONS ??



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THANK YOU



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